Chicago Crime Patterns and Socioeconomic Correlates

Group Members	Xinying Jiang, Henry Liu	**Lecture Section**	Section 4	**GitHub**	xinying0301, henryliu1125

Research Question

This project investigates two related questions about crime in Chicago in 2024: (1) *What temporal and spatial patterns characterize crime across Chicago census tracts?* and (2) *How do neighborhood-level socioeconomic conditions — education attainment, unemployment, and median income — relate to local crime rates?*

The primary data source is the City of Chicago's publicly available crime incident dataset (2001–present), filtered to 2024, which provides 256,652 geocoded records across 862 census tracts. Socioeconomic variables are drawn from the 2024 American Community Survey 5-Year Estimates (ACS): bachelor's degree attainment rate (B15003), unemployment rate (S2301), and median household income (B19013). A 2023 TIGER/Line census tract shapefile was used to join all datasets geographically.

Approach and Coding

The analysis was organized in four stages. First, **data cleaning**: the 31 original crime primary types were collapsed into four categories (Violent, Property, Regulatory, Other) using a custom classification function. ACS tables required skipping metadata rows, extracting 11-digit GEOID tract codes, and computing derived rates (e.g., bachelor-plus rate = sum of bachelor's through doctorate holders divided by total population 25+).

Second, **spatial joining**: a GeoPandas `sjoin` with predicate `"within"` assigned each crime point to its census tract. The city boundary shapefile clipped tracts to Chicago, removing partial Cook County tracts at the city edge. Crime counts were then aggregated by tract and a per-100-resident crime rate was computed.

Third, **static visualization**: eight Altair charts were produced — two temporal heatmaps, two choropleth maps of crime count and rate, a bar chart of crime by type, and three socioeconomic choropleth maps. Quantile color scales were used to account for skewed distributions.

Fourth, a **Streamlit app** was built to allow interactive exploration of crime patterns and their relationship to socioeconomic variables.

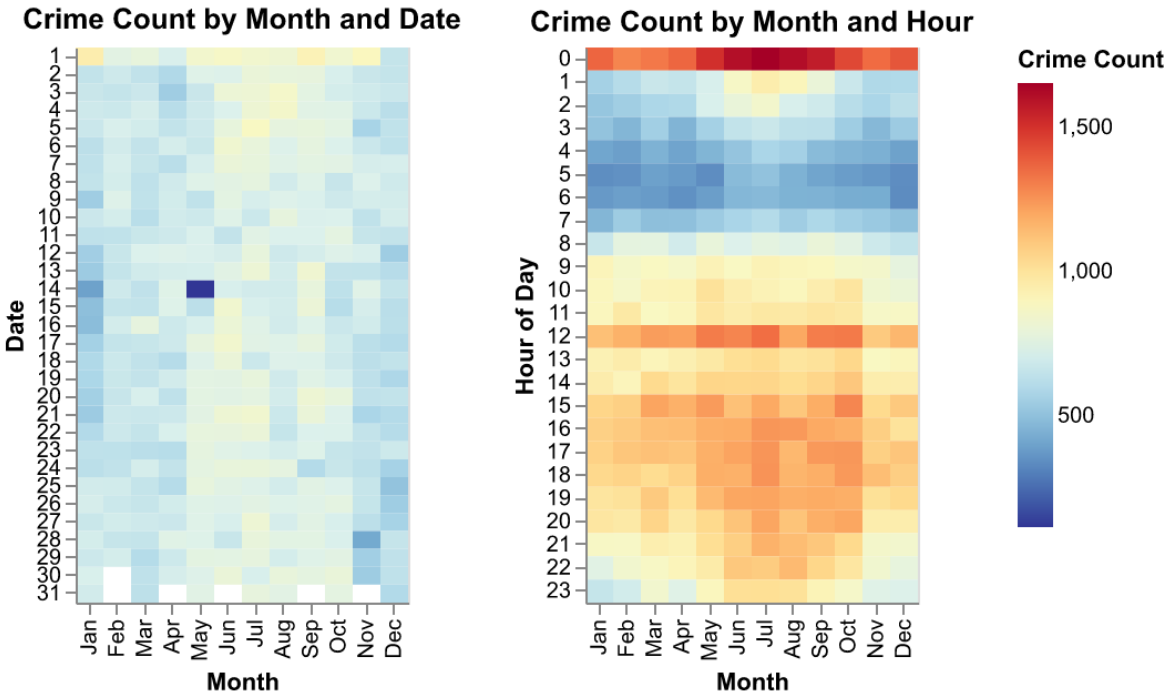
Weaknesses and difficulties encountered:

- *Spatial join performance*: Joining ~259,000 crime points against 862 tract polygons was computationally intensive; ensuring a shared CRS (EPSG:4326) before joining was necessary to avoid silent mismatches.
- *ACS metadata rows*: ACS CSVs contain two non-data header rows, requiring careful row-skipping (`iloc[3:]`) and explicit numeric conversion.
- *Choropleth rendering*: Altair's max-rows limit required disabling the data transformer for GeoJSON-heavy maps; PNG export via `vl-convert` was used to ensure readable resolution.
- *Cross-sectional limitation*: Both crime and ACS data are from 2024, so analysis is correlational only. Some high-rate tracts have small residential populations (e.g., airport or park areas), artificially inflating per-capita rates.

Static Plots

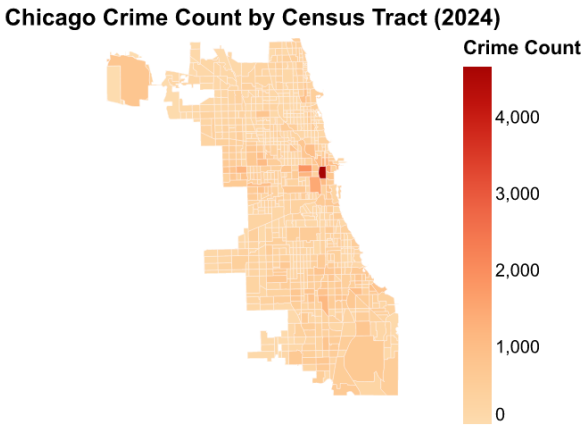
Plots 1 & 2 — Crime Count by Month/Date and Month/Hour

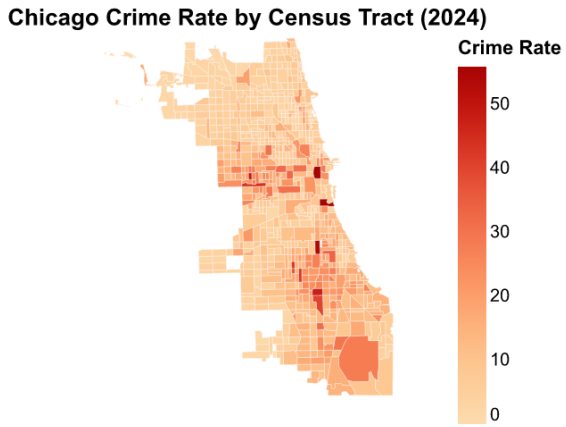
The heatmaps reveal clear temporal structure. Plot 1 (Month × Date) shows higher crime in summer months and on the 1st of each month. Plot 2 (Month × Hour) shows a midday peak around noon and an evening peak (6–9 PM), especially in summer. Early-morning hours (2–5 AM) are uniformly low. Together these plots show that crime follows both seasonal and daily rhythms, directly informing when and where intervention resources should be concentrated.



Plots 3 & 4 — Crime Count and Crime Rate Choropleth Maps

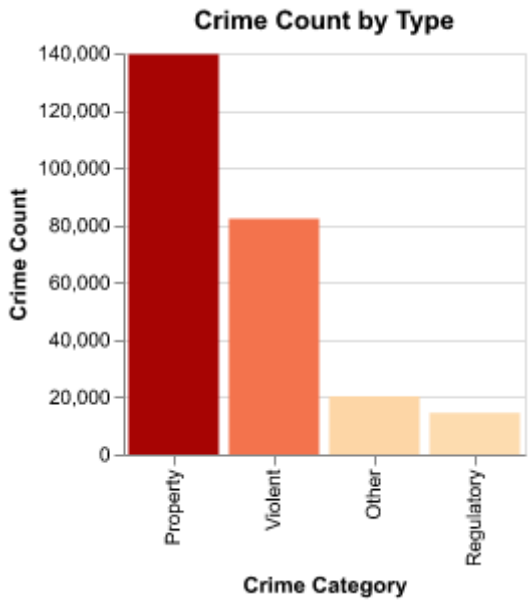
Plot 3 maps raw crime counts, with the highest concentration in the Loop and Near North Side (dense commercial and transit areas). Plot 4 adjusts for population (per 100 residents), revealing that several South and West Side tracts carry the highest crime *rates* even where absolute counts are moderate — pointing to neighborhoods with elevated community-level risk rather than land-use intensity.

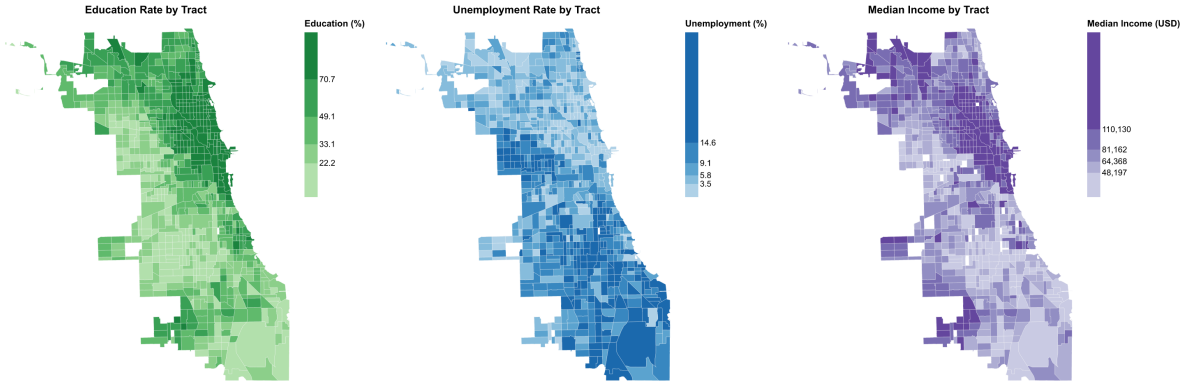




Plot 5 — Crime Count by Type

Property crime dominates (~140,000 incidents), followed by Violent (~80,000), Other (~22,000), and Regulatory (~18,000). The prevalence of economically motivated property crime suggests it is more likely to co-vary with income and unemployment, motivating the socioeconomic analysis in Part 2.



Plots 6, 7 & 8 — Socioeconomic Choropleth Maps

The three maps provide spatial context for interpreting crime rates. Higher education (Plot 6, green) concentrates on the North Side; lower values mirror high-crime South/West Side tracts. Unemployment (Plot 7, blue) clusters in many of the same South/West Side tracts with elevated crime rates. Median income (Plot 8, purple) reinforces this pattern. Visually comparing these to Plot 4 reveals a consistent spatial correlation between socioeconomic disadvantage and crime burden.